

Affine Transforms, Three Covariances, and Borel Clusters of an Exponential Hawkes Process

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Abstract

For the exponential Hawkes process, we close the joint count–intensity transform, the stationary intensity law, and a triangular recurrence for all moments. The subcritical condition is treated together with its parameter faces. All results are source-local; no arithmetic or target spectral data enter.

1 The predictable Markov owner

Let $b > 0$, $a, \nu \geq 0$, and let N be a simple point process whose predictable event rate is the left limit λ_{t-} , where

$$d\lambda_t = -b(\lambda_t - \nu) dt + a dN_t. \quad (1)$$

Thus an event is sampled with the pre-jump intensity, then raises the post-jump intensity by a . Its Markov generator is

$$\mathcal{G}f(x) = b(\nu - x)f'(x) + x\{f(x + a) - f(x)\}. \quad (2)$$

This convention prevents the common but consequential confusion between λ_{t-} and λ_t at an event.

Theorem 1 (joint affine transform). *For $0 \leq z \leq 1$, $s \geq 0$, and $\lambda_0 = x$,*

$$\mathbb{E}_x[z^{N_t - N_0} e^{-s\lambda_t}] = e^{-A_t - B_t x}, \quad (3)$$

where $B_0 = s$, $A_0 = 0$, and

$$B'_t = 1 - bB_t - ze^{-aB_t}, \quad A'_t = b\nu B_t. \quad (4)$$

Proof. In the backward equation, a jump multiplies the transform by z . Substituting $F(t, x) = e^{-A_t - B_t x}$ into (2), with $f(x + a)$ replaced by $zf(x + a)$, gives constant coefficient $-b\nu B_t$ and x -coefficient $bB_t + ze^{-aB_t} - 1$. Equating these with $-A'_t - B'_t x$ proves (4). The transform is bounded on the stated parameter domain, so localization and bounded convergence close the calculation. $\square$

2 Stationarity, Laplace law, and every moment

Suppose first that $a < b$, set $\delta = b - a$, and put $\mu = b\nu/\delta$. The excitation kernel is $h(t) = ae^{-bt}\mathbf{1}_{t>0}$ and has mass $a/b < 1$.

Theorem 2 (stationary closure). *For $\nu > 0$ there is a unique finite-intensity stationary Hawkes law. Its intensity Laplace transform $L(s) = \mathbb{E}e^{-s\lambda}$ is the unique normalized solution*

$$\frac{L'(s)}{L(s)} = -\frac{b\nu s}{bs + e^{-as} - 1}, \quad L(0) = 1. \quad (5)$$

Writing $m_n = \mathbb{E}\lambda^n$ and $m_0 = 1$, every moment is finite and

$$m_n = \frac{nb\nu m_{n-1} + \sum_{k=0}^{n-2} \binom{n}{k} a^{n-k} m_{k+1}}{n\delta}, \quad n \geq 1. \quad (6)$$

In particular, $\mathbb{E}\lambda = \mu$ and $\text{Var } \lambda = \mu a^2/(2\delta)$.

Proof. The immigration–Poisson-cluster construction is subcritical, hence locally finite and stationary; the same genealogy gives uniqueness among finite-intensity stationary Hawkes processes. Applying stationary (2) to e^{-sx} gives

$$0 = -b\nu sL(s) - \{bs + e^{-as} - 1\}L'(s).$$

The bracket has derivative $b - ae^{-as} > 0$ and vanishes only at zero, so (5) has one normalized solution. Applying (2) to x^n , expanding $x\{(x+a)^n - x^n\}$, and isolating the coefficient $n(b-a)m_n$ gives (6). Induction closes every moment, and its first two instances give the displayed mean and variance. $\square$

References

- [1] A. G. Hawkes, *Spectra of some self-exciting and mutually exciting point processes*, Biometrika **58** (1971), 83–90, [doi:10.1093/biomet/58.1.83](https://doi.org/10.1093/biomet/58.1.83).